

Spheroid, basement membrane, matrix:

the three-entity model and the arrows between them

the container note · prototype/ecm · log/okuda_ECM · companions: note_junction, note_fibre, note_surface,
note_sheet · 10 August 2026

1. Mechanism

An epithelial spheroid growing inside an extracellular matrix is *three* entities and not two, and the middle one is not a passive spacer. The literature fixes its proportions, and they are not the convenient ones.

The basement membrane is thin, stiff and squat-anchored. It is ~ 100 nm thick; its adhesions are clusters, not molecules: an integrin is a heterodimer, integrins assemble into nanoclusters of 20–50 at ~ 35 –100 nm [14], and those cluster again into adhesions with a measured ~ 555 nm centre-to-centre spacing [15]. The **plaque** set is the MIDDLE level. Vertically the cluster sits ~ 40 nm from the actin it pulls on [16], so an adhesion is several times *wider* than the sheet is thick and a fraction of its own spacing across. And the sheet is 10 to 100 times stiffer than the stroma outside it, not equal to it.

It is not a permanent solid. Collagen IV in the fly embryo has a half-life of 7–10 hours [13] and in mouse hair follicle about 3, and slowing that turnover disrupts morphogenesis. A sheet enclosing a tissue that triples its radius does not stretch to 2.4; it is *replaced* at the new size while it is being stretched.

And it is the interface. While the sheet is intact the epithelium does not touch the stroma: load goes cell $\rightarrow$ plaque $\rightarrow$ sheet $\rightarrow$ anchoring fibril $\rightarrow$ stroma [17]. An epithelium that *also* pushes the stroma directly is counting the same push twice. That single observation is what removes **cell_to_ecm** from the model and puts two edge sets in its place.

2. Equations and symbols

Each equation below belongs to one arrow of Fig. 1b. The companion notes carry the mechanics of each entity; this note carries what passes *between* them.

Mass, not just mechanics. The sheet carries an areal density ρ of collagen IV produced by the epithelium and removed by proteolysis, on a surface whose area is growing:

$$\frac{D\rho}{Dt} = \underbrace{s(\mathbf{x}, t)}_{\text{secretion}} - \underbrace{\frac{\rho}{\tau_{\text{bm}}}}_{\text{turnover}} - \underbrace{\rho \frac{\dot{A}}{A}}_{\text{dilution}}, \quad s^* = \rho \left(\frac{1}{\tau_{\text{bm}}} + \frac{\dot{A}}{A} \right), \quad \tau_{\text{bm}} = \frac{t_{1/2}}{\ln 2} \approx 4\text{--}14 \text{ h}. \quad (1)$$

The two removal terms are the same size here — a spheroid tripling its radius over a day has $\dot{A}/A \sim 10^{-1} \text{ h}^{-1}$ against $1/\tau_{\text{bm}} \sim 10^{-1} \text{ h}^{-1}$. Turnover is not a correction to growth; it is the same size as growth, which is why the sheet can accommodate it at all.

A nonlinear sheet. A single Young’s modulus is wrong by construction: the same membrane reads ~ 0.4 MPa at low strain and ~ 3 MPa at high [12], and a growing spheroid takes it across that range:

$$E(\lambda) = E_0 [1 + \beta(\lambda - 1)], \quad \lambda = \sigma_{\text{max}}(\mathbf{F}), \quad \frac{E_{\text{bm}}}{E_{\text{ecm}}} \in [10, 100]. \quad (2)$$

MPa is not a quantity this box has — it is dimensionless and the stroma’s modulus is the number 15 — so the falsifiable forms are the two ratios and never the pascals.

Adhesion, with its reaction and with slip. A plaque is two-sided: whatever force it applies to the sheet it applies equally and oppositely to the cell, which a replayed epithelium discards. And the sheet is not pinned — it *slides* — so the tangential law is friction against relative velocity and not a spring to a frozen direction:

$$\mathbf{f}_p^{\text{adh}} = \kappa_n (\ell_p - \ell_0) \hat{\mathbf{n}}_p - \xi (\mathbf{v}_{\text{bm}} - \mathbf{v}_{\text{epi}})_{\parallel}, \quad \mathbf{f}_p^{\text{epi}} = -\mathbf{f}_p^{\text{adh}}, \quad (3)$$

$$n_p = \Sigma^{-2} \text{ plaques per unit area, } \Sigma \approx 7T, \quad \ell_0 \approx 0.3T. \quad (4)$$

ℓ_0 is a material property and not a tuning constant: it is what sets the standoff.

Sheet to stroma. Anchoring fibrils [17] load the stroma *through* the sheet, which is also what lets a stiff sheet spread a local epithelial push over a soft matrix:

$$\mathbf{t} = k_a (\mathbf{x}_{\text{ecm}} - \mathbf{x}_{\text{bm}}) \text{ on anchored points, } \mathbf{f}^{\text{bm}} = - \sum_{\text{anchors}} \mathbf{t}. \quad (5)$$

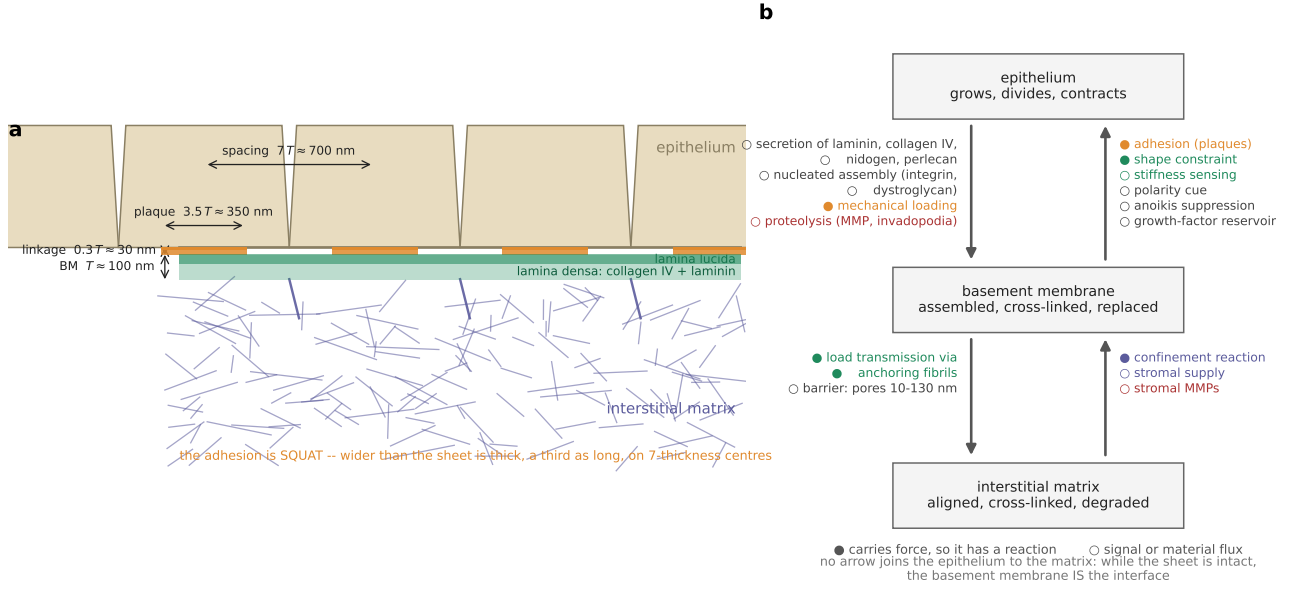


Figure 1: (a) The three entities in section, at the proportions the literature measures rather than the ones that are convenient: the adhesion is a squat plaque, not a thread, and the sheet is thin against its own spacing. (b) Every operator, in the direction it acts. Filled bullets carry a force and must therefore have a reaction; open bullets carry a signal or a material flux. There is no arrow between the epithelium and the matrix.

The barrier is geometric rather than mechanical: pores of 10–130 nm pass solutes, stop cells [11], and are what a proteolytic operator has to open before anything crosses.

The matrix pushes back on the cell cycle. The force the stroma returns, collected by direction and divided by the area each direction covers, is a pressure; a cell facing a stressed matrix grows more slowly, which is what a spheroid grown under a measured external stress does [9, 10]:

$$g = f + \frac{1 - f}{1 + (P/p_{1/2})^n}, \quad s_{\text{now}} \leftarrow s_{\text{prev}} (1 + g (f_{\text{grow}} - 1)). \quad (6)$$

Because it is the growth *rate* that is slowed and not the size directly, a small difference compounds over hundreds of frames — which is how a modest difference in stress between one direction and another becomes a difference in shape.

symbol	is	of what
$\rho, \tau_{\text{bm}}, s$	areal density, turnover time, secretion	of collagen IV on the sheet, Eq. (1)
$A, \dot{A}/A$	sheet area, its relative growth rate	the dilution term; the SAME size as turnover
E_0, β, λ	low-strain modulus, stiffening, principal stretch	of the sheet, Eq. (2); report RATIOS, never MPa
T, Σ, ℓ_0	thickness, plaque spacing, linkage rest length	$\Sigma \approx 7T, \ell_0 \approx 0.3T$; T never appears as geometry
κ_n, ξ	plaque normal stiffness, tangential friction	against RELATIVE velocity — ξ is what lets the sheet stay attached while the tissue slides
k_a	fibril stiffness	sheet to stroma, Eq. (5)
$P, p_{1/2}, n, f$	matrix pressure, gate half-point, sharpness, floor	Eq. (6); $f \leq g \leq 1$ multiplies the growth RATE

3. The Plexus2 container

Two structural claims, and both differ from what the archived line built. An adhesion is not a body and a fibril is not a particle: each is a *relation between two entities*, which in Plexus2 is an edge set carrying `.pre` and `.post` index buffers rather than a set of nodes with positions. Writing the hemidesmosome as an edge set is what makes its reaction automatic — one operator returns a delta to both endpoints, so the force the sheet feels and the force the cell feels cannot get out of step. The archived line made the adhesion first a field on the membrane and then a column of MPM particles, and both lost the reaction.

set	kind	one element is	state it carries
spheroid	node, depth 0	the organ	centroid, cell count
cell	node, depth 1	one epithelial cell	A, P, V and targets, age, ρ_f
vertex	node, depth 1	a corner of the apical mesh	pos
junction	edge $\subseteq$ vertex ²	a cell-cell contact	n_e, ℓ_e
bm	node on a surface mesh	a patch of membrane	pos , F , ρ , age, occ
bm_face	hyperedge $\subseteq$ bm ³	a triangle of membrane	$\mathbf{D}_m^{-1}$, $\mathbf{C}_0$, A^0 , Y_2
plaque	edge cell $\rightarrow$ bm	one hemidesmosome	ℓ_0 , load, bound
fibril	edge bm $\rightarrow$ mpm_particle	one anchoring fibril	rest length, load
mpm_particle	node, depth 1	a patch of <i>stroma only</i>	pos , F , fibre direction
mpm_grid	field	the stroma's momentum	mass, momentum
protease	field	MMP concentration	scalar, diffusing

operator	kind	acts on	mechanism	where its gates are
<i>epithelium</i>				
grow_3d, divide_3d	Divide	cell	volume growth and cytokinesis [6, 7]	00; note_junction
shape_energy_3d	Lateral	junction	vertex-model force balance [8]	00
medioapical_myosin,	Lateral,	cell,	two myosin pools and the ring	note_junction J1–
junction_myosin,	Structural	junction		J11
cytokinetic_ring				
<i>epithelium</i> $\rightarrow$ <i>basement membrane</i>				
bm_secrete,	Divide, Die	bm_face	Eq. (1) [13, 18]	note_sheet G24–
bm_degrade				G26, G19–G23
bm_stiffness,	Lateral	bm_face	Eq. (2)	note_sheet G1–G8
bm_elastic				
<i>basement membrane</i> $\leftrightarrow$ <i>epithelium</i>				
plaque_seed,	Rewire,	plaque	Eqs. (3)–(4), <i>with the reaction</i> [16]	note_sheet G9–
plaque_pull,	Lateral			G13
plaque_rupture				
<i>basement membrane</i> $\leftrightarrow$ <i>matrix</i>				
fibril_seed,	Rewire,	fibril	Eq. (5) [17]	open
fibril_pull	Lateral			
mesh_contact	Lateral	contact	particle-to-surface, both endpoints in one call	note_surface S1–
				S19
bm_barrier	Exchange	bm_face, fields	pores pass solutes, stop cells [11]	open
<i>matrix</i>				
seed_ecm,	mpm_*, Seed, Ex-	mpm_particle	MLS-MPM [4, 5]	note_fibre F1–F18
ecm_stress	change, Lateral			
ecm_growth_gate_3d	Lateral	cell	Eq. (6) [9, 10]	58; open

One schedule, split at exactly one place. Anything carrying a force runs inside the substep block, at the stroma's rate; anything changing topology or mass runs once per frame, because a set whose membership changed mid-substep would be integrated against two different index sets in one step:

frame ($\Delta t = 1$): cell_geometry_3d $\rightarrow$ grow_3d $\rightarrow$ bm_growth_gate $\rightarrow$ medioapical_myosin $\rightarrow$ junction_myosin
 $\rightarrow$ shape_energy_3d $\rightarrow$ reconnect_t1_3d $\rightarrow$ divide_3d $\rightarrow$ cytokinetic_ring $\rightarrow$ junction_myosin
 $\rightarrow$ bm_refine $\rightarrow$ bm_secrete $\rightarrow$ bm_degrade $\rightarrow$ plaque_seed $\rightarrow$ plaque_rupture

substep ($\Delta t_{\text{sub}} = \Delta t/8$): bm_stiffness $\rightarrow$ bm_elastic $\rightarrow$ plaque_pull $\rightarrow$ fibril_pull
 $\rightarrow$ mpm_strain $\rightarrow$ mpm_scatter
 $\rightarrow$ mpm_grid_update $\rightarrow$ mpm_gather

Four rules, each buying something specific. *The membrane never enters the grid* — it is ~ 100 nm on a $\sim 200 \mu\text{m}$ spheroid, so resolving it as MPM material needs a 3500^3 lattice; as a surface mesh it costs what the epithelium costs and Δx is set by the stroma alone. *The epithelium stays out of the substep block* — it is overdamped and massless, so it does not need the matrix's CFL step, and putting its relaxation inside would multiply its cost by eight and change nothing. *The sheet is massless too*, so its stiffness costs drag rather than substeps: the bound is $\Delta t (zk_b + \kappa)/\gamma < 2$, a rate and not a wave speed, and the 10–100 stiffness ratio is paid for with γ . *Particles go on the shell that carries stress, not on the box* — the far field is quiescent, and filling the box at a usable density costs a million particles.

And one thing this deletes. cell_to_ecm and cell_exclude_3d go, because while the sheet is intact the basement membrane *is* the interface (§1). MPM fibres also stop being asked to act as ropes, which note_fibre measured they cannot.

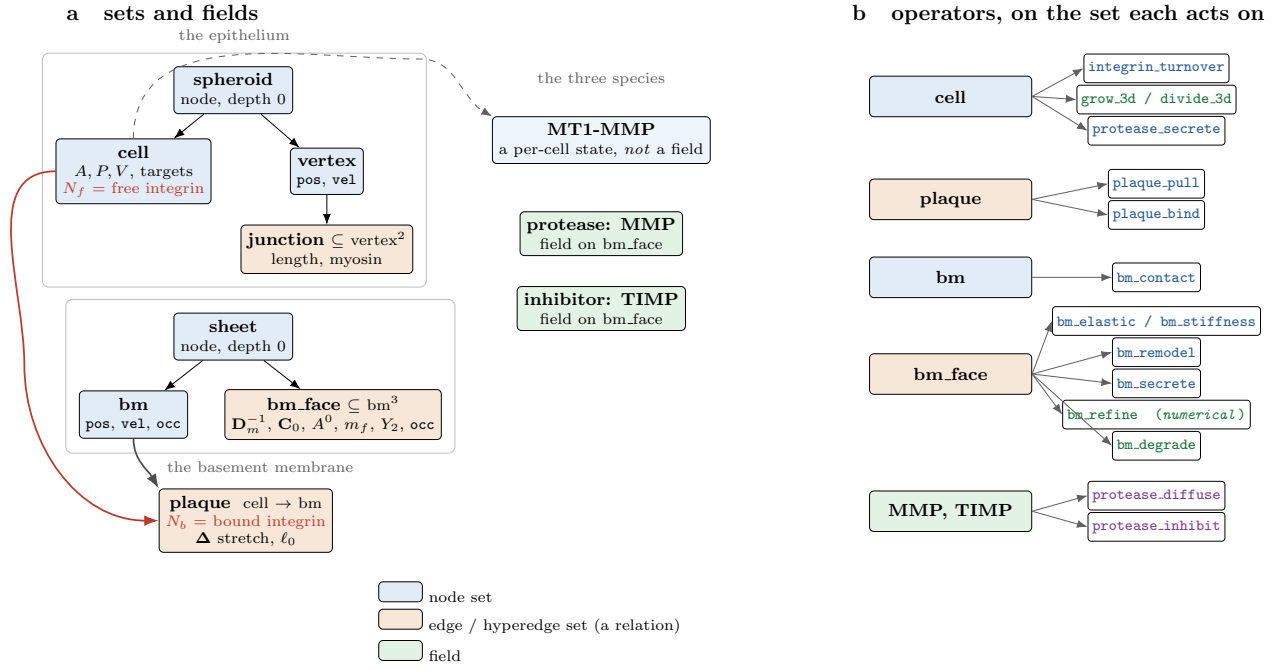


Figure 2: **(a)** Every set and field, and which kind each is. A node set carries positions; an edge or hyperedge set is a *relation* and carries state belonging to neither endpoint — a junction to neither cell, a plaque to neither the cell nor the sheet, a triangle to none of its three nodes. **(b)** Every operator, on the set it acts on, coloured by kind. Anything carrying a *force* sits on a set whose elements have positions; anything carrying *mass* or *number* sits on the set that owns the quantity. **bm_refine** is the only operator here that is numerical rather than biological.

3.1 The two diffusible species, and the one that is not

Proteolysis introduces the first *fields* in this model, and deciding which of the three species deserves to be one is not a modelling preference — it is arithmetic.

species	what it is	where it lives	why
MMP	soluble MMP-2/9, ~ 72 kDa, cuts collagen IV	field on bm_face	it diffuses freely, $D \sim 10\text{--}100 \mu\text{m}^2/\text{s}$
TIMP	soluble inhibitor, binds MMP 1:1	field on bm_face	likewise — and it is the <i>sink</i>
MT1-MMP	membrane-type MMP14, transmembrane	a per-cell <i>state</i>	it never enters the extracellular space

A soluble protease cannot open a hole, and the arithmetic says so before any run. In one 600 s frame a soluble MMP travels $\sqrt{4Dt} = 155\text{--}490 \mu\text{m}$, against a spheroid $318 \mu\text{m}$ across and a cell of $10 \mu\text{m}$. It is therefore *uniform over the entire sheet within a single frame*, and a uniform protease dissolves rather than perforates — the same failure uniform starvation produced, for the same reason. Localising to one cell would need $D < 0.042 \mu\text{m}^2/\text{s}$, $240\text{--}2400\times$ slower than any protein of that size. That is not a parameter range, it is a different molecule, and the different molecule is MT1-MMP: *tethered*, acting where its cell put it. This is why focal basement-membrane breach is done by membrane-type MMPs and invadopodia rather than by soluble enzyme.

What makes the soluble field worth solving at all is the sink. With $Dc/Dt = D\nabla_s^2 c + s - kc$ the source decays over $\sqrt{D/k}$, so the field acquires a *length* instead of filling the sphere — and k here is not a fitted decay but the reaction with TIMP. The breach diameter becomes a *prediction* of the MMP/TIMP balance: a $10 \mu\text{m}$ breach needs an effective k of $0.1\text{--}0.5 \text{ s}^{-1}$ (half-life $1.4\text{--}6.9$ s), a $50 \mu\text{m}$ breach $0.004\text{--}0.02 \text{ s}^{-1}$ ($35\text{--}170$ s). A number to check, not to choose.

Three new operators, and one costs what the others do not. **protease_secrete** (Lateral, on **cell**) is the source; **protease_inhibit** (Exchange) is the 1:1 reaction; **protease_diffuse** (Exchange) is the only operator in this model carrying a *spatial* coupling, and so the only one whose stability limit involves the mesh. Explicit diffusion needs $\Delta t D/h^2 < \frac{1}{2}$ — at these D , $1,300\text{--}13,400$ substeps per frame, *quadrupling at every 1 $\rightarrow$ 4 refinement*, unlike the sheet's own bound which is a rate and mesh-independent. It is therefore solved semi-implicitly, $(\mathbf{I} + \Delta t D\mathbf{L})c = c_{\text{old}}$ by conjugate gradient on a finite-volume face Laplacian: unconditionally stable, one solve per frame, indifferent to h .

The same argument applied to the integrin gives the opposite answer for the opposite reason. A receptor is well mixed because it is *confined* to one cell's basal face ($\sqrt{4Dt} = 15.5 \mu\text{m}$ over a $10 \mu\text{m}$ cell) and needs

no field; a soluble protease would be well mixed because it *travels far*. **Confinement, not diffusivity, decides whether a species is worth resolving as a field.**

Proteolysis, and the mechanism the literature forced

What the proteases are for, before any of their own chemistry. Everything up to here builds a barrier: a sheet that carries load, holds its density, and stays attached. The proteases are the only operators in the model that *remove* it, and they do so *locally* — so they are the mechanism of **barrier breach**, the step by which an epithelium stops being enclosed and a cell can cross into the stroma. Their target is the sheet’s material (**bm_degrade** acts on **bm_face** mass); adhesion is affected only downstream, because a bond needs ligand, so ρ falling takes n_b with it.

05g_degrade put the three species on the entities the confinement argument assigns them and measured which one can make a hole. The answer was unambiguous and one-sided:

arm	faces dead	under MT1	elsewhere	sheet breached
both arms	317	317	0	6.2%
soluble only	0	0	0	0%
tethered only	317	317	0	6.2%
no protease	0	0	0	0%
no inhibitor	5120	317	4803	100%

The last row is the argument of §3.1 as an experiment: **remove TIMP and the sheet dissolves entirely**, with 4803 of the deaths nowhere near a cell that expresses MT1-MMP. With the inhibitor present the diffusible arm kills nothing at all and the tethered arm kills exactly its own footprint. A soluble protease has no length scale unless something quenches it.

Two defects that the runs found rather than the design. First, neither field had a *clearance* term, so TIMP accumulated without bound (0.0318 at frame 20, 0.0438 at frame 25) — its only sink was a 1:1 reaction with an MMP twenty times scarcer, so there was no steady state for the pair to reach. A first-order clearance is not a refinement: the $\sqrt{D/k}$ that justifies solving the field *is* that k . With it, both fields plateau and drift by $< 10^{-4}$ over twenty frames. Second, a five-point sweep of the MMP:TIMP source ratio over a hundredfold range returned *identically* 317 dead at every point, because the direct tethered cut had been set $350\times$ larger than anything the soluble arm could contribute — the sweep axis had no effect on the answer, and the holes were the source stencil in every case.

TIMP-2 is the adaptor as well as the inhibitor, and that predicts a bell

The pure-inhibitor law of 05g is half of the mechanism. In the canonical model [1] TIMP-2 is the *bridging adaptor* that makes activation possible: its C-terminus tethers proMMP-2 and its N-terminus binds MT1-MMP, and the prodomain is then cleaved by a *second*, *TIMP-2-free* MT1-MMP [2]. TIMP-2 therefore both presents the zymogen and poisons the enzyme that must cut it, and the dose response is biphasic. The dissection is clean: an artificial receptor that tethers proMMP-2 *without* inhibiting MT1-MMP has no descending limb [3], so the fall-off is inhibition and not saturation.

With T the tethered enzyme and K the MT1–TIMP dissociation constant, occupancy is Langmuir and the rate needs one occupied site and one free one:

$$[\text{MT1}\cdot\text{TIMP}] = T \frac{c_T/K}{1 + c_T/K}, \quad [\text{MT1}_{\text{free}}] = \frac{T}{1 + c_T/K}, \quad \text{act} \propto T^2 c_{\text{pro}} \frac{c_T/K}{(1 + c_T/K)^2}, \quad (7)$$

which is bell-shaped and **peaks at** $c_T = K$. Both the shape and the location are therefore predictions, and 05g’s pure-inhibitor law cannot produce a peak at all — it can only decay — so one sweep distinguishes them and 05g is the control.

Measured (05h_ternary, eight TIMP levels, four species): the activation rate rises and falls, $4.5 \times 10^{-2} \rightarrow 6.2 \times 10^{-2} \rightarrow 2.4 \times 10^{-2} \rightarrow 7.5 \times 10^{-3} \rightarrow 2.5 \times 10^{-3}$, and the pure-inhibitor control over the same sweep is monotone decreasing, as Eq. (7) requires of it. Two arrangements had to be corrected before the curve could be read at all, and both are worth recording because they are measurement errors rather than modelling ones: the activation had to be made *rate-limiting* (at a large k_{act} the whole zymogen pool converts every frame whatever TIMP does, and saturation erases the dose response); and the rate has to be read *while the substrate still exists*, since at the TIMP levels where the mechanism works best it destroys every MT1-bearing face and a late-window average then reports exactly zero at the peak.

4. Gates

A gate is a number with a threshold decided *before* the run and a stated consequence if it fails. The per-entity gates live in the companion notes and are indexed in §3; this table is the *system* level — what has to be true of the assembled model, and which arrows are not yet built.

gate	threshold	measured
<i>the epithelium is unchanged by everything attached to it</i> (00 is the reference)		
B1 cells, radius, aspect, T1 rate of the bare tissue	the numbers every later run must re-produce bit-identical	200 → 6076, × 3.56 , 1.021 , 0.0089 /cell/frame
B2 adding per-junction myosin at $m_e \equiv 1$ changes nothing	bit-identical	bit-identical
<i>each arrow carries its reaction</i>		
B3 epithelium → matrix, momentum residual	$< 10^{-6}$	1.5–1.8e-7 across every sweep run (note_surface S5, S12)
B4 cell ↔ sheet, as bookkeeping and as motion	$< 10\epsilon_{64}$; drift $< 10^{-12}$	1.4e-16 , 4.4e-15 (note_sheet G9, G10)
B5 sheet → stroma: the stroma is loaded THROUGH the sheet	displacement with the sheet present vs absent	not built — fibril_pull is the missing arrow
<i>the middle entity does what a membrane does</i>		
B6 the sheet reports its own deformation	> 0.95 of the geometric stretch within 10%	0.9935 as a surface; 0.13 as MPM material (run 130)
B7 ρ held flat while A triples	within 10%	1/11.1 with no secretion (note_sheet G24)
B8 the standoff is the linkage's rest length	$\ell/\ell_0 \in [0.9, 1.1]$	3.43 (note_sheet G12)
<i>the coupling is mutual</i>		
B9 the tissue feels the matrix: shape differs from the free control	any separation oblateness against the uncon-fined run	one-way — the reaction is computed, recorded and dropped
B10 the growth gate changes the shape it is given	oblateness against the uncon-fined run	1.43 grown vs 1.41 pressed (58, 48)

One convention makes these mean what they say. A reaction is measured on its own force pair, inside the substep and before any drive is added; with the drive in the sum it stops being a statement about the reaction. That is why B3 and B4 are separate rows with different thresholds: B3 is float32 through a grid, B4 is float64 on an explicit edge.

What the gates license. The epithelium is reproducible on its own and nothing attached to it has moved it (B1, B2); every arrow that has been built returns its reaction to machine precision (B3, B4); the sheet, as a surface, reports the deformation it is given (B6); and the growth gate can produce a shape that pressing produces too, which is the one place the matrix has been shown to change the tissue (B10). **What they do not license** is the assembled model: B5 is not built, B7 and B8 fail on the sheet's own testbench, and B9 is the architecture — the tissue is a replay, so it drives and is never driven. Until B9 passes, the stroma's modulus is very nearly a free parameter: the tissue never feels the matrix, and the growth gate normalises the pressure map to its own late-run median, so a softer stroma changes the substep it needs and not the shape that comes out. That is the clearest statement of what one-way coupling costs, and **mesh_contact** returning **vertex_force** to a live epithelium is the run that ends it.

What sixty-six archived runs settled, and it is four things. Runs 91–164 asked where the sheet sits and what holds it there, and they answer that question and no other. The standoff is a *tracking lag*, $\delta - \dot{R}\Delta t\gamma/\kappa$, and not a force balance, which two fibre lengths differing fourfold confirmed to 3×10^{-5} box units. A force routed through the grid leaves the sheet carrying 100% of its true stretch while the same force integrated by the engine leaves it carrying 13% (B6). An MPM fibre cannot act as a rope, because $\mathbf{F}$ is advected by the grid's velocity gradient and never measures the distance between two particles. And every MPM body in the prototype silently carried the *parent* set's modulus, so a membrane declaring $E = 400$ ran at the stroma's 15 until each body was given its own parent — which is why run 146 came back bit-identical to run 144 at ten times the stiffness. None of the four is a model derived from what a basement membrane *is*; that is what **note_sheet** builds.

5. References

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